INCH-POUND

MIL-C-7115/3B(AS)
26 August 1991

SUPERSEDING
MIL-C-7115/3A(AS)
15 March 1991

MILITARY SPECIFICATION SHEET

CONVERTER, AIRCRAFT, 150 AMPERE TYPE II, BLAST COOLED, CLASS C

This specification is approved for use by the Naval Air Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the converter described herein shall consist of this document and the latest issue of MIL-C-7115.

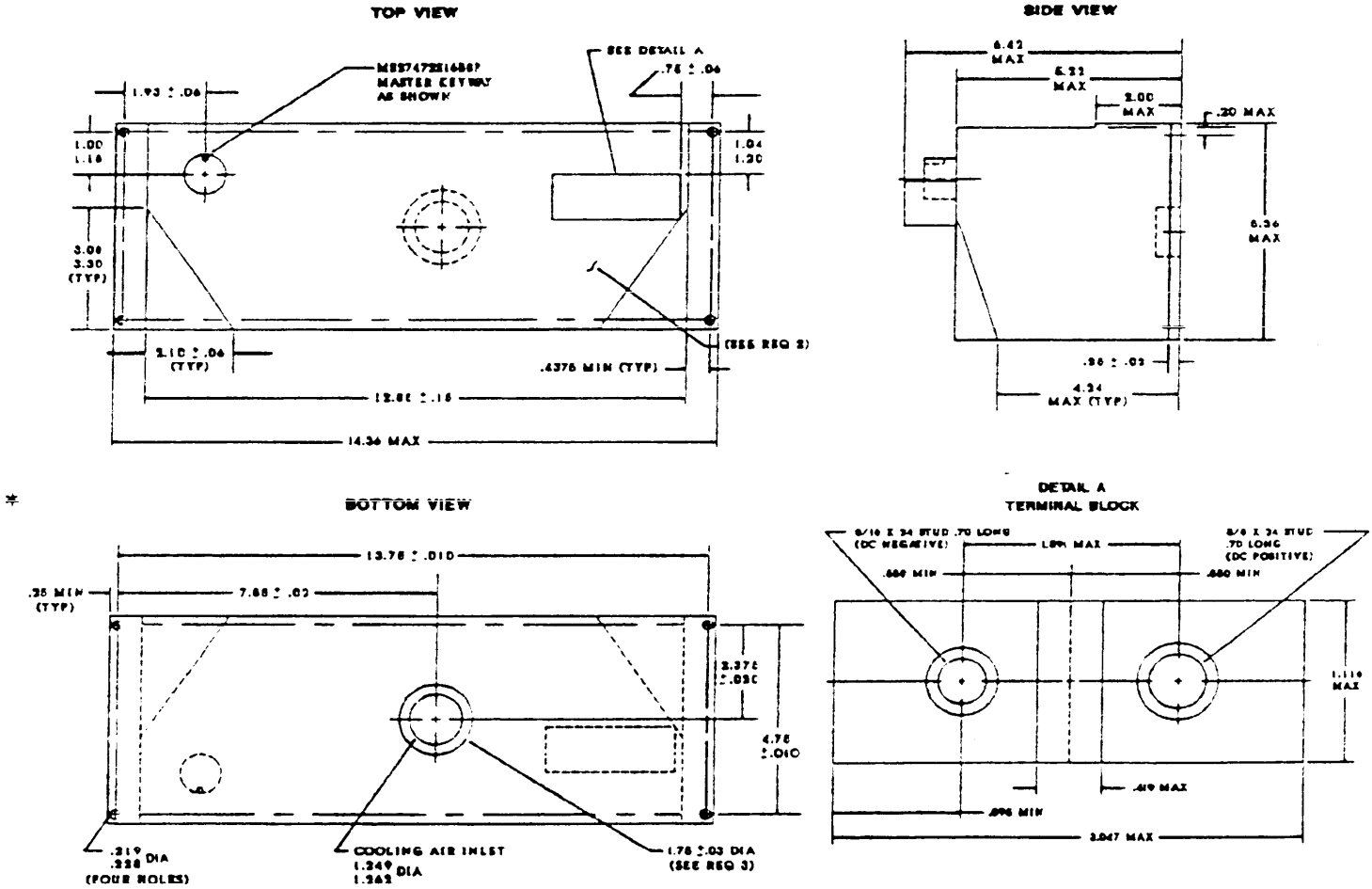


FIGURE 1. Converter P/N M7115/3-1.

AMSC N/A FSC 6130

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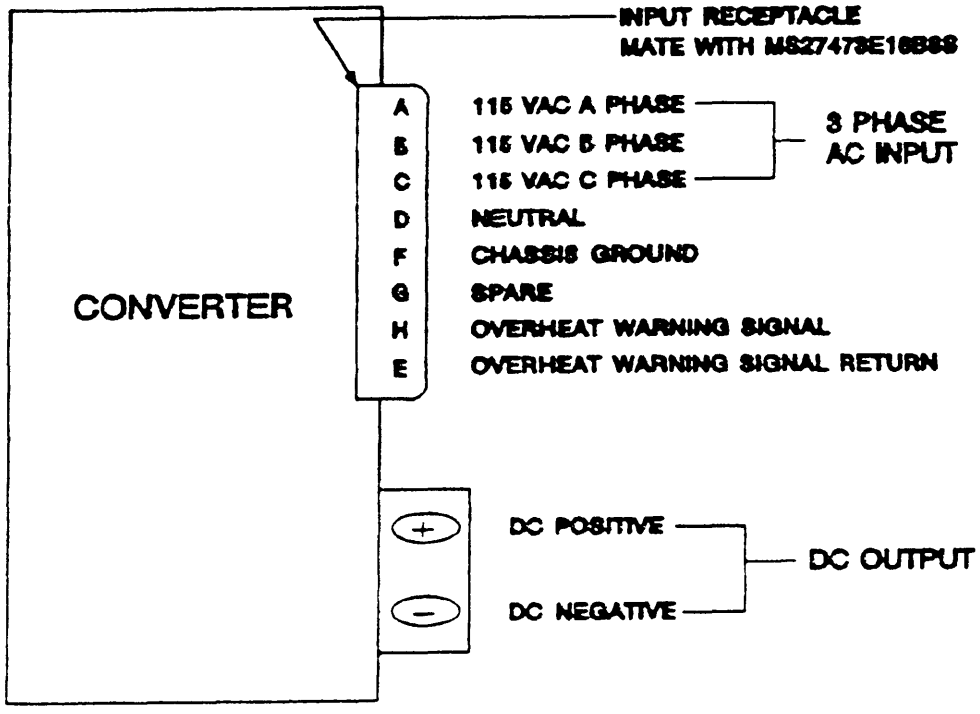


FIGURE 2. Connection diagram.

TABLE I.

AC INPUT:	
Voltage Range	112.5 - 117.5 Volts RMS
Frequency Range	380 - 420 Hertz
DC OUTPUT:	
Voltage Range	See Figure 5
Current Range	1.5 - 150 Amperes
Inlet Airflow vs Temperature Range:	See Figure 3
Temperature/Altitude Range:	See Figure 4
Weight:	14.7 lbs (max)

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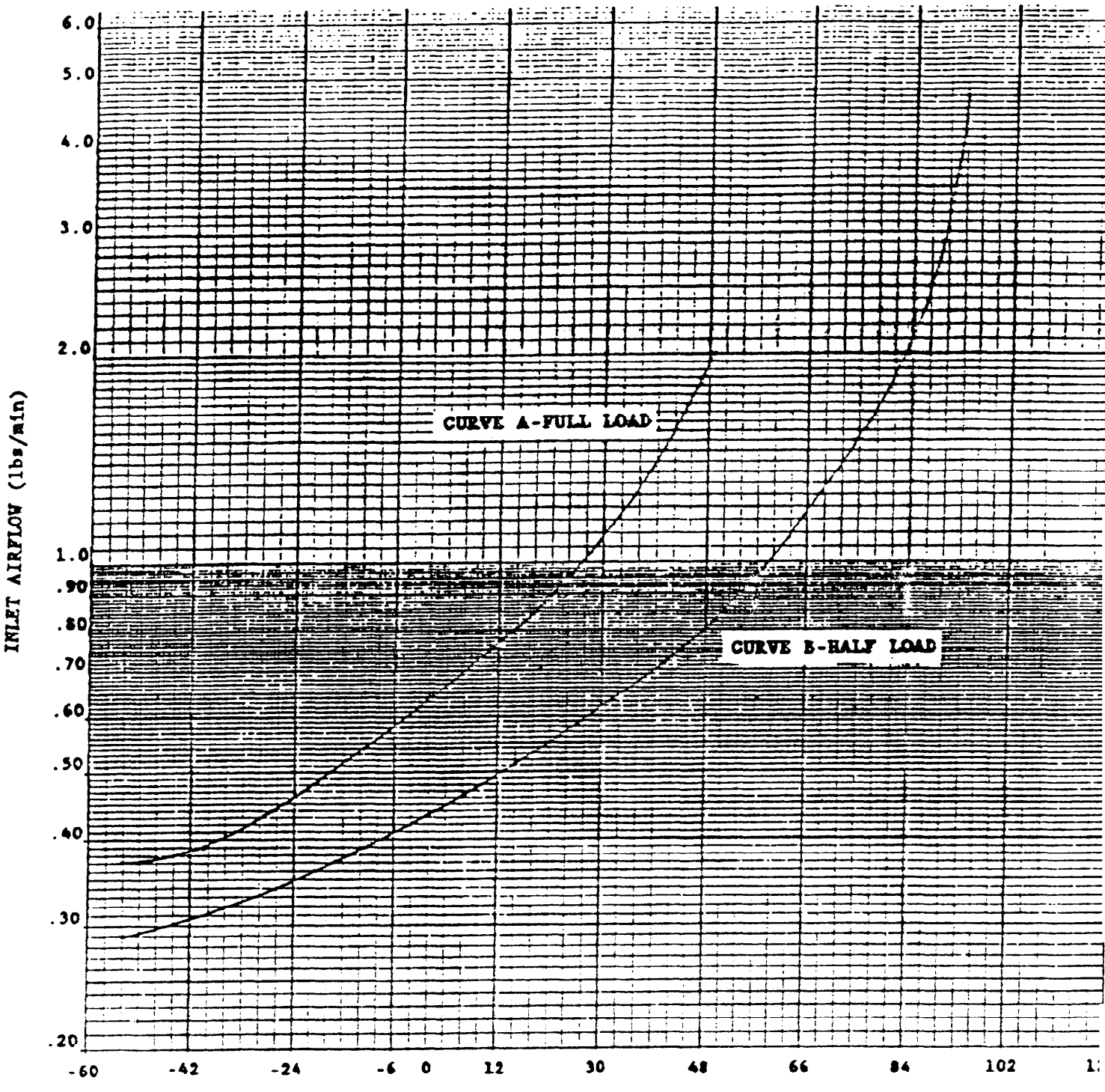


FIGURE 3-INLET AIRFLOW VS INLET TEMPERATURE FOR CONTINUOUS OPERATION

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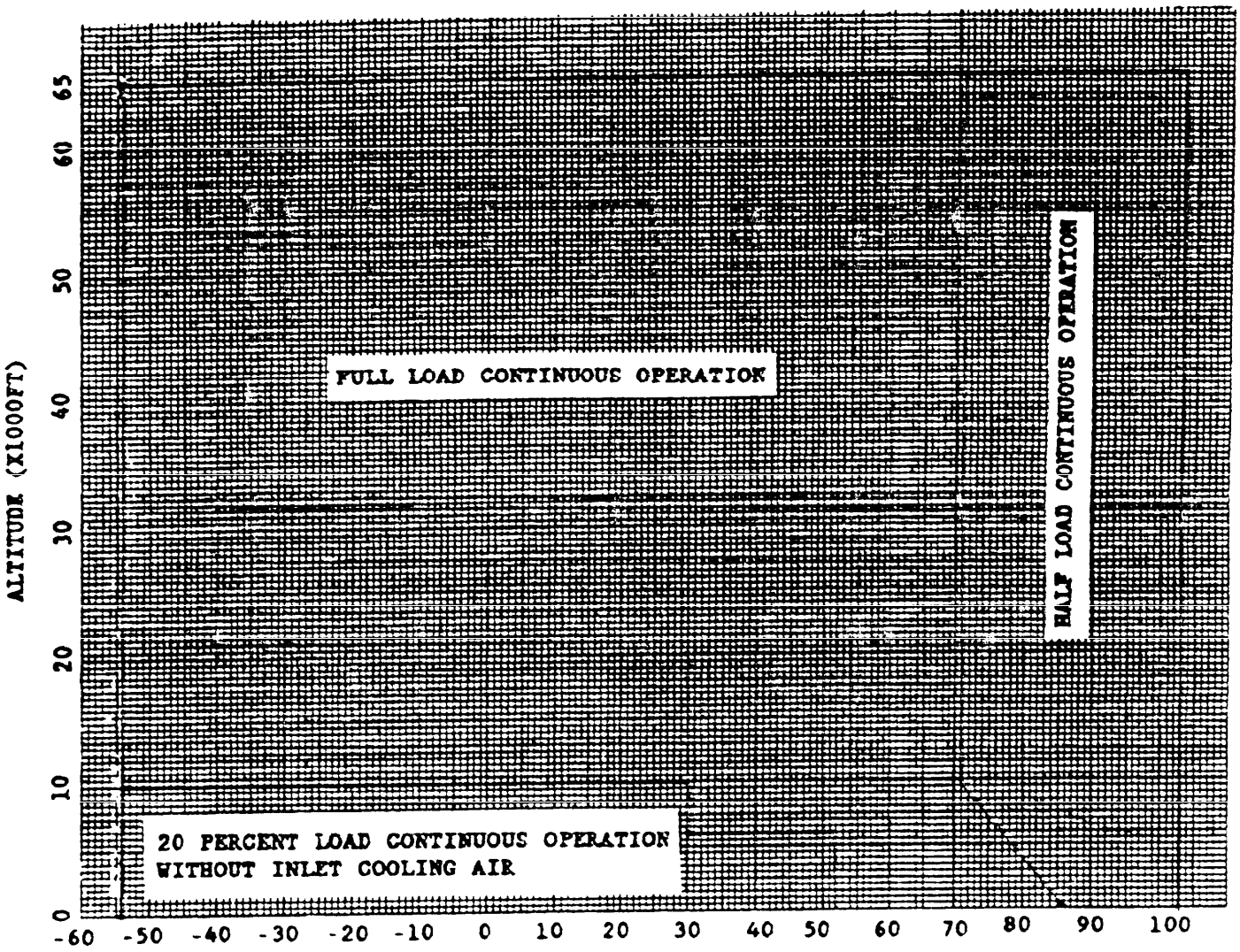
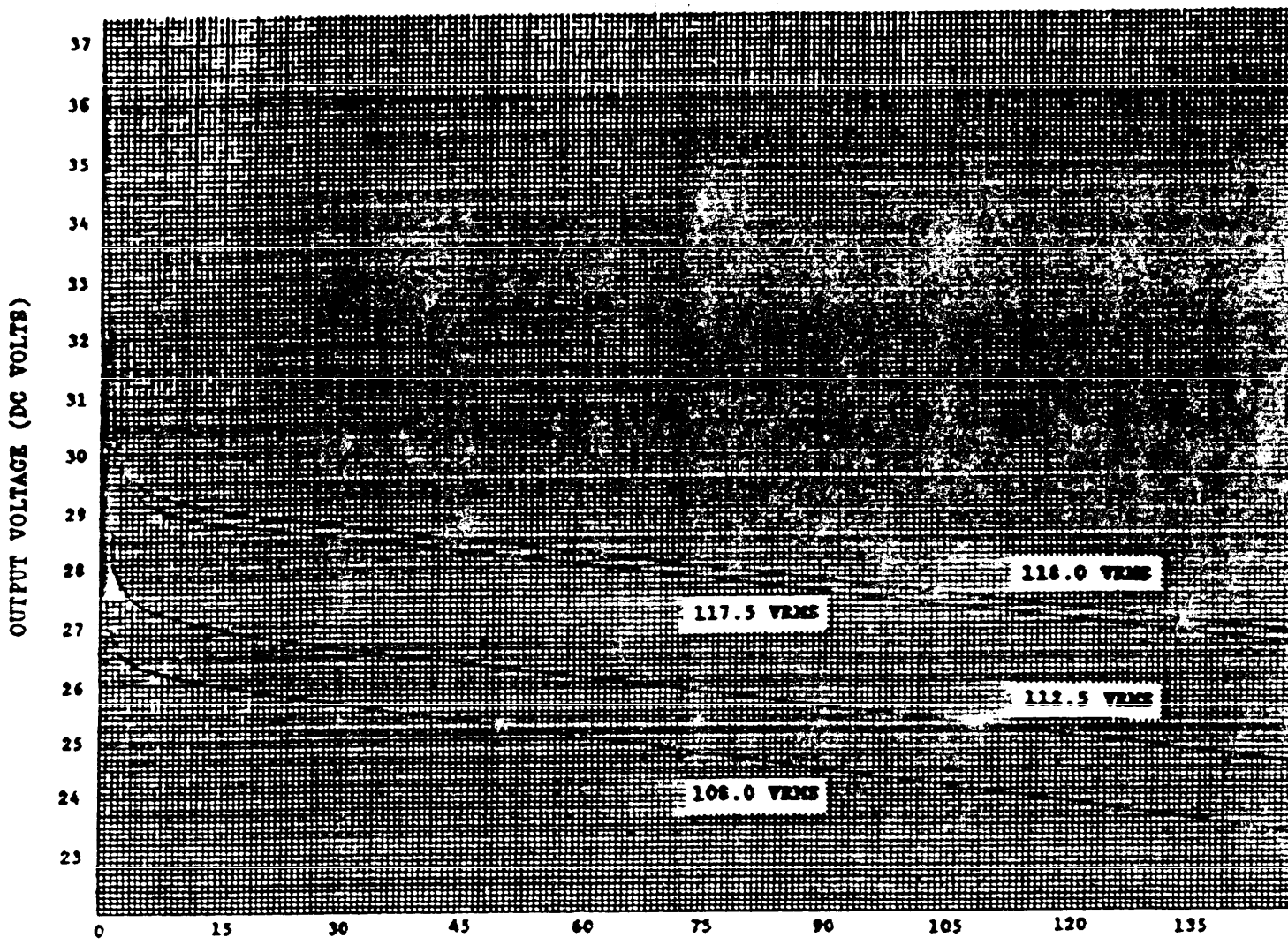


FIGURE 4. TEMPERATURE/ALTITUDE

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INPUT		108.0 VRMS	112.5 VRMS	117.5 VRMS	118.0 VRMS
OUTPUT	AMPS DC	MIN. VDC	MIN. VDC	MAX. VDC	MAX. VDC
	*0.0	31.0	32.2	36.3	36.7
	1.5	26.7	27.6	30.3	30.5
	5.0	26.3	27.4	29.2	29.4
	15.0	26.0	27.1	28.8	29.0
	30.0	25.7	26.7	28.6	28.8
	37.5	25.5	26.6	28.4	28.6
	75.0	24.7	25.9	27.8	28.0
	112.5	24.1	25.2	27.2	27.4
	150.0	23.3	24.5	26.6	26.8

FIGURE 5-OUTPUT VOLTAGE LIMITS

*No load data provided for information only

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REQUIREMENTS:

1. Blast air shall exit from the top side of the converter where the terminal block and connector are located.

2. The nameplate and caution placard, which states "Caution Hot Surface" shall be located on the top side of the converter. A cover stiffener is permitted on the side of the converter where the terminal block and connector are located. The cover stiffener shall not exceed 0.27 inches (max) above the surface.

3. The surface within this diameter is required for a sealing gasket. It shall be free of burrs and defects of ± 0.015 inches or greater, and shall be in the same plane as the four mounting holes.

4. The converter shall meet the requirements of MIL-C-7115 as modified below: (paragraph numbers refer to MIL-C-7115)

3.4.1.3.3 Color. Delete "color number 17875 (white)" of first sentence and add "color number 37038 (black)."

3.5.2 Non-standard parts. Add "In the event that a non-standard part was approved during the qualification process of the converter, it is not necessary to request approval for use of the same non-standard part in production converters."

3.6.1.1 Component derating. Delete and add "All converter electric/electronic components shall be derated as shown in Table I of MIL-C-7115 based upon the most severe converte environmental and operational conditions except as modified in Table II of this specification sheet below:

TABLE II.

Part Type	Critical Stress	Maximum Allowable Application Stress
Diodes Rectifier	PIV	.60
Capacitors, Fixed Plastic	Voltage	.67
Miscellaneous Transformers and Chokes	Flux Saturation level Current	.95 .90

3.6.2.6 Cooling. Add "The converter shall meet all the requirements of this specification when supplied the minimum inlet airflow of Figure 3 of this specification sheet which will be delivered to the converter through the cooling air inlet at the various ambient temperature - altitude conditions of Figure 4 of this specification sheet. Maximum inlet airflow is 5.0 lb/min."

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3.6.2.7 Ventilation openings. Add "except for the cooling air inlet, which shall be fr of screening or obstruction."

Add the following new paragraph:

3.6.2.10 Overheat warning. A thermal switch in accordance with MIL-S-24236 shall b provided within the converter to sense when an overheat condition exists. The thermal switch sha be normally open, and shall close at the onset of an overheat condition. The switch shall not clo during any specified operating condition, and shall contain no electronic parts or semiconducto devices. The converter shall not be degraded by verification of operation test per paragra 4.7.2.29, and shall then pass the test of paragraph 4.7.2.17.

3.7.1.1.5 Input current balance. Delete "5 percent" and add "10 percent".

Add the following new paragraph:

3.7.1.1.7 Ungrounded neutral operation. The three-phase input power neutral shall not b internally grounded to the chassis of the converter. The converter shall meet the requirements o this specification when operated with the three-phase input power neutral lead grounded o ungrounded (disconnected).

3.7.1.2.1.1aDelete and add "Steady state output voltage shall remain within the limits o Figure 5 of this specification sheet."

3.7.1.2.1.1bDelete and add "Peak to mean ripple voltage for either polarity shall not excee 1.75 Volts from 0 to 10 percent rated load current, and 1.5 volts from 10 percent to 100 percen rated load current."

3.7.1.2.2 Shock load recovery. Delete and add "The output voltage transient limits of Figur 2 of MIL-C-7115 shall apply except for the first 2.5 milliseconds from the onset of the transient The following transient limits shall apply for the first 2.5 milliseconds from the onset of th transient:

Condition	Output Transient Limit (Volts)
Application of 100% load current	15 (min)
Application of 200% load current	9 (min)
Removal of 100% load current	37.5 (max)

3.7.1.2.3 Output starting voltage. Add to the second sentence "except for the first seconds from the onset of the transient, when a maximum of 45 Volts shall be allowed."

* 3.7.1.4.2 Type II nonregulated converters. Add "except the conducted emission requirements of CE03, MIL-STD-461 shall be relaxed 5 db from 150 KHz to 2 Mhz for th output leads and the input neutral lead."

3.7.1.5 Acoustical noise. Not applicable.

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3.7.1.6 Pressure drop (blast-cooled converters only). Add "The pressure drop through the converter shall not exceed the following:

0.3 inches of water at 17°C, 0.52 lb/min and 14.7 psia ambient.

36.0 inches of water at 103°C, 5.0 lb/min and 14.7 psia ambient.

The blast air supplied may contain entrained (free) moisture up to 104 grains per pound of air."

3.7.2.1 Temperature-altitude. Delete and add "The converter shall meet all performance requirements specified herein throughout the temperature-altitude range of Figure 4 of this specification sheet."

3.7.2.2 Acceleration. Not applicable.

3.7.2.9 Explosive atmosphere. Not applicable.

3.9.3 Nameplate. Add "A caution placard shall be afixed to the top side of the converter which states: Caution Hot Surface."

4.7.1.5 Airflow. Delete last sentence and add "Nominal airflow is 0.69 lb/min at 5°C."

4.7.2.2.1a Delete last sentence and add "For blast-cooled converters only during this portion of the test at half and full rated load, the input airflow conditions shall be varied from nominal conditions throughout the entire range of Figure 3 of this specification sheet, to establish the airflow conditions which result in maximum and then minimum internal component temperatures of the converter when stabilized at half and full rated load. Additionally, maximum and minimum internal component temperatures shall be determined at 20 percent load current and no inlet airflow."

4.7.2.2.2 Voltage regulation at maximum temperature-altitude conditions. Add "The converter shall be tested under the temperature-altitude conditions as shown in Figure 4 of this specification sheet. The maximum ambient temperature shall be as specified in Figure 4 of this specification sheet."

4.7.2.2.3 Voltage regulation at minimum temperature-altitude conditions. Delete "-30°C" and replace with "-55°C".

4.7.2.10 Overload and short circuit capacity. Delete last sentence and add "The highest current test point shall be 750 percent of the maximum rated load, and shall be conducted for the corresponding duration indicated in Figure 4."

4.7.2.15 Acoustical noise. Not applicable.

4.7.2.18 Acceleration. Not applicable.

4.7.2.21 Vibration. Delete first sentence and add "The converter shall be tested in accordance with MIL-E-81910 except that both random and sinusoidal vibration shall be conducted on one converter."

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4.7.2.25 Explosive atmosphere. Not applicable.

* 4.7.2.29 Overheat warning. The converter shall be stabilized under nominal conditions. The inlet airflow shall then be removed. The converter shall be deenergized when the thermal switch closes.

NOTES:

1. Dimensions are in inches. Tolerance is ±0.015 unless otherwise specified.

2. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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	MIL-C-7115/3B(AS)	910830

3. DOCUMENT TITLE

CONVERTER, AIRCRAFT, 150 AMPERE, TYPE II, BLAST COOLED, CLASS C

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

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Ken Bladen (AIR-53633E) Electrical Power Section	(703) 692-2653	222-2653
c. ADDRESS (Include Zip Code) Commander Naval Air Systems Command Washington, D.C. 20361-5360	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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